

Therefore, the Eq. (17) becomes:

$$\begin{aligned}
& \frac{1}{\sqrt{(2\pi)^d |\Sigma|}} \exp\left(-\frac{(x - \mu_1)^\top \Sigma^{-1} (x - \mu_1)}{2}\right) \pi_1 \\
&= \frac{1}{\sqrt{(2\pi)^d |\Sigma|}} \exp\left(-\frac{(x - \mu_2)^\top \Sigma^{-1} (x - \mu_2)}{2}\right) \pi_2, \\
&\Rightarrow \exp\left(-\frac{(x - \mu_1)^\top \Sigma^{-1} (x - \mu_1)}{2}\right) \pi_1 \\
&= \exp\left(-\frac{(x - \mu_2)^\top \Sigma^{-1} (x - \mu_2)}{2}\right) \pi_2, \\
&\stackrel{(a)}{\Rightarrow} -\frac{1}{2}(x - \mu_1)^\top \Sigma^{-1} (x - \mu_1) + \ln(\pi_1) \\
&= -\frac{1}{2}(x - \mu_2)^\top \Sigma^{-1} (x - \mu_2) + \ln(\pi_2),
\end{aligned}$$

where (a) takes natural logarithm from the sides of equation.

We can simplify this term as:

$$\begin{aligned}
& (x - \mu_1)^\top \Sigma^{-1} (x - \mu_1) = (x^\top - \mu_1^\top) \Sigma^{-1} (x - \mu_1) \\
&= x^\top \Sigma^{-1} x - x^\top \Sigma^{-1} \mu_1 - \mu_1^\top \Sigma^{-1} x + \mu_1^\top \Sigma^{-1} \mu_1 \\
&\stackrel{(a)}{=} x^\top \Sigma^{-1} x + \mu_1^\top \Sigma^{-1} \mu_1 - 2 \mu_1^\top \Sigma^{-1} x, \quad (19)
\end{aligned}$$

where (a) is because $x^\top \Sigma^{-1} \mu_1 = \mu_1^\top \Sigma^{-1} x$ as it is a scalar and Σ^{-1} is symmetric so $\Sigma^{-\top} = \Sigma^{-1}$. Thus, we have:

$$\begin{aligned}
& -\frac{1}{2} x^\top \Sigma^{-1} x - \frac{1}{2} \mu_1^\top \Sigma^{-1} \mu_1 + \mu_1^\top \Sigma^{-1} x + \ln(\pi_1) \\
&= -\frac{1}{2} x^\top \Sigma^{-1} x - \frac{1}{2} \mu_2^\top \Sigma^{-1} \mu_2 + \mu_2^\top \Sigma^{-1} x + \ln(\pi_2).
\end{aligned}$$

Therefore, if we multiply the sides of equation by 2, we have:

$$\begin{aligned}
& 2 (\Sigma^{-1} (\mu_2 - \mu_1))^\top x \\
&+ (\mu_1 - \mu_2)^\top \Sigma^{-1} (\mu_1 - \mu_2) + 2 \ln\left(\frac{\pi_2}{\pi_1}\right) = 0, \quad (20)
\end{aligned}$$

which is the equation of a line in the form of $a^\top x + b = 0$. Therefore, if we consider Gaussian distributions for the two classes where the covariance matrices are assumed to be equal, the decision boundary of classification is a line. Because of linearity of the decision boundary which discriminates the two classes, this method is named *linear discriminant analysis*.

For obtaining Eq. (20), we brought the expressions to the right side which was corresponding to the second class; therefore, if we use $\delta(x) : \mathbb{R}^d \rightarrow \mathbb{R}$ as the left-hand-side expression (function) in Eq. (20):

$$\begin{aligned}
\delta(x) &:= 2 (\Sigma^{-1} (\mu_2 - \mu_1))^\top x \\
&+ (\mu_1 - \mu_2)^\top \Sigma^{-1} (\mu_1 - \mu_2) + 2 \ln\left(\frac{\pi_2}{\pi_1}\right), \quad (21)
\end{aligned}$$

the class of an instance x is estimated as:

$$\hat{C}(x) = \begin{cases} 1, & \text{if } \delta(x) < 0, \\ 2, & \text{if } \delta(x) > 0. \end{cases} \quad (22)$$

If the priors of two classes are equal, i.e., $\pi_1 = \pi_2$, the Eq. (20) becomes:

$$\begin{aligned}
& 2 (\Sigma^{-1} (\mu_2 - \mu_1))^\top x \\
&+ (\mu_1 - \mu_2)^\top \Sigma^{-1} (\mu_1 - \mu_2) = 0, \quad (23)
\end{aligned}$$

whose left-hand-side expression can be considered as $\delta(x)$ in Eq. (22).

4. Quadratic Discriminant Analysis for Binary Classification

In Quadratic Discriminant Analysis (QDA), we relax the assumption of equality of the covariance matrices:

$$\Sigma_1 \neq \Sigma_2, \quad (24)$$

which means the covariances are not *necessarily* equal (if they are actually equal, the decision boundary will be linear and QDA reduces to LDA).

Therefore, the Eq. (17) becomes:

$$\begin{aligned}
& \frac{1}{\sqrt{(2\pi)^d |\Sigma_1|}} \exp\left(-\frac{(x - \mu_1)^\top \Sigma_1^{-1} (x - \mu_1)}{2}\right) \pi_1 \\
&= \frac{1}{\sqrt{(2\pi)^d |\Sigma_2|}} \exp\left(-\frac{(x - \mu_2)^\top \Sigma_2^{-1} (x - \mu_2)}{2}\right) \pi_2, \\
&\stackrel{(a)}{\Rightarrow} -\frac{d}{2} \ln(2\pi) - \frac{1}{2} \ln(|\Sigma_1|) \\
&\quad - \frac{1}{2} (x - \mu_1)^\top \Sigma_1^{-1} (x - \mu_1) + \ln(\pi_1) \\
&= -\frac{d}{2} \ln(2\pi) - \frac{1}{2} \ln(|\Sigma_2|) \\
&\quad - \frac{1}{2} (x - \mu_2)^\top \Sigma_2^{-1} (x - \mu_2) + \ln(\pi_2),
\end{aligned}$$

where (a) takes natural logarithm from the sides of equation. According to Eq. (19), we have:

$$\begin{aligned}
& -\frac{1}{2} \ln(|\Sigma_1|) - \frac{1}{2} x^\top \Sigma_1^{-1} x - \frac{1}{2} \mu_1^\top \Sigma_1^{-1} \mu_1 \\
&+ \mu_1^\top \Sigma_1^{-1} x + \ln(\pi_1) \\
&= -\frac{1}{2} \ln(|\Sigma_2|) - \frac{1}{2} x^\top \Sigma_2^{-1} x - \frac{1}{2} \mu_2^\top \Sigma_2^{-1} \mu_2 \\
&+ \mu_2^\top \Sigma_2^{-1} x + \ln(\pi_2).
\end{aligned}$$

Therefore, if we multiply the sides of equation by 2, we have:

$$\begin{aligned}
& x^\top (\Sigma_1 - \Sigma_2)^{-1} x + 2 (\Sigma_2^{-1} \mu_2 - \Sigma_1^{-1} \mu_1)^\top x \\
&+ (\mu_1^\top \Sigma_1^{-1} \mu_1 - \mu_2^\top \Sigma_2^{-1} \mu_2) + \ln\left(\frac{|\Sigma_1|}{|\Sigma_2|}\right) \\
&+ 2 \ln\left(\frac{\pi_2}{\pi_1}\right) = 0, \quad (25)
\end{aligned}$$

which is in the quadratic form $\mathbf{x}^\top \mathbf{A} \mathbf{x} + \mathbf{b}^\top \mathbf{x} + c = 0$. Therefore, if we consider Gaussian distributions for the two classes, the decision boundary of classification is quadratic. Because of quadratic decision boundary which discriminates the two classes, this method is named *quadratic discriminant analysis*.

For obtaining Eq. (25), we brought the expressions to the right side which was corresponding to the second class; therefore, if we use $\delta(\mathbf{x}) : \mathbb{R}^d \rightarrow \mathbb{R}$ as the left-hand-side expression (function) in Eq. (25):

$$\begin{aligned} \delta(\mathbf{x}) := & \mathbf{x}^\top (\Sigma_1 - \Sigma_2)^{-1} \mathbf{x} + 2 (\Sigma_2^{-1} \mu_2 - \Sigma_1^{-1} \mu_1)^\top \mathbf{x} \\ & + (\mu_1^\top \Sigma_1^{-1} \mu_1 - \mu_2^\top \Sigma_2^{-1} \mu_2) + \ln\left(\frac{|\Sigma_1|}{|\Sigma_2|}\right) + 2 \ln\left(\frac{\pi_2}{\pi_1}\right), \end{aligned} \quad (26)$$

the class of an instance $\mathbf{x}$ is estimated as the Eq. (22).

If the priors of two classes are equal, i.e., $\pi_1 = \pi_2$, the Eq. (20) becomes:

$$\begin{aligned} & \mathbf{x}^\top (\Sigma_1 - \Sigma_2)^{-1} \mathbf{x} + 2 (\Sigma_2^{-1} \mu_2 - \Sigma_1^{-1} \mu_1)^\top \mathbf{x} \\ & + (\mu_1^\top \Sigma_1^{-1} \mu_1 - \mu_2^\top \Sigma_2^{-1} \mu_2) + \ln\left(\frac{|\Sigma_1|}{|\Sigma_2|}\right) = 0, \end{aligned} \quad (27)$$

whose left-hand-side expression can be considered as $\delta(\mathbf{x})$ in Eq. (22).

5. LDA and QDA for Multi-class Classification

Now we consider multiple classes, which can be more than two, indexed by $k \in \{1, \dots, |\mathcal{C}|\}$. Recall Eq. (12) or (15) where we are using the scaled posterior, i.e., $f_k(\mathbf{x}) \pi_k$. According to Eq. (16), we have:

$$\begin{aligned} & f_k(\mathbf{x}) \pi_k \\ &= \frac{1}{\sqrt{(2\pi)^d |\Sigma_k|}} \exp\left(-\frac{(\mathbf{x} - \mu_k)^\top \Sigma_k^{-1} (\mathbf{x} - \mu_k)}{2}\right) \pi_k. \end{aligned}$$

Taking natural logarithm gives:

$$\begin{aligned} \ln(f_k(\mathbf{x}) \pi_k) = & -\frac{d}{2} \ln(2\pi) - \frac{1}{2} \ln(|\Sigma_k|) \\ & - \frac{1}{2} (\mathbf{x} - \mu_k)^\top \Sigma_k^{-1} (\mathbf{x} - \mu_k) + \ln(\pi_k). \end{aligned}$$

We drop the constant term $-(d/2) \ln(2\pi)$ which is the same for all classes (note that this term is multiplied before taking the logarithm). Thus, the scaled posterior of the k -th class becomes:

$$\begin{aligned} \delta_k(\mathbf{x}) := & -\frac{1}{2} \ln(|\Sigma_k|) \\ & - \frac{1}{2} (\mathbf{x} - \mu_k)^\top \Sigma_k^{-1} (\mathbf{x} - \mu_k) + \ln(\pi_k). \end{aligned} \quad (28)$$

In QDA, the class of the instance $\mathbf{x}$ is estimated as:

$$\hat{\mathcal{C}}(\mathbf{x}) = \arg \max_k \delta_k(\mathbf{x}), \quad (29)$$

because it maximizes the posterior of that class. In this expression, $\delta(\mathbf{x})$ is Eq. (28).

In LDA, we assume that the covariance matrices of the k classes are equal:

$$\Sigma_1 = \dots = \Sigma_{|\mathcal{C}|} = \Sigma. \quad (30)$$

Therefore, the Eq. (28) becomes:

$$\begin{aligned} \delta_k(\mathbf{x}) = & -\frac{1}{2} \ln(|\Sigma|) \\ & - \frac{1}{2} (\mathbf{x} - \mu_k)^\top \Sigma^{-1} (\mathbf{x} - \mu_k) + \ln(\pi_k) = -\frac{1}{2} \ln(|\Sigma|) \\ & - \frac{1}{2} \mathbf{x}^\top \Sigma^{-1} \mathbf{x} - \frac{1}{2} \mu_k^\top \Sigma^{-1} \mu_k + \mu_k^\top \Sigma^{-1} \mathbf{x} + \ln(\pi_k). \end{aligned}$$

We drop the constant terms $-(1/2) \ln(|\Sigma|)$ and $-(1/2) \mathbf{x}^\top \Sigma^{-1} \mathbf{x}$ which are the same for all classes (note that before taking the logarithm, the term $-(1/2) \ln(|\Sigma|)$ is multiplied and the term $-(1/2) \mathbf{x}^\top \Sigma^{-1} \mathbf{x}$ is multiplied as an exponential term). Thus, the scaled posterior of the k -th class becomes:

$$\delta_k(\mathbf{x}) := \mu_k^\top \Sigma^{-1} \mathbf{x} - \frac{1}{2} \mu_k^\top \Sigma^{-1} \mu_k + \ln(\pi_k). \quad (31)$$

In LDA, the class of the instance $\mathbf{x}$ is determined by Eq. (29), where $\delta(\mathbf{x})$ is Eq. (31), because it maximizes the posterior of that class.

In conclusion, QDA and LDA deal with maximizing the *posterior* of classes but work with the *likelihoods* (class conditional) and *priors*.

6. Estimation of Parameters in LDA and QDA

In LDA and QDA, we have several parameters which are required in order to calculate the posteriors. These parameters are the means and the covariance matrices of classes and the priors of classes.

The priors of the classes are very tricky to calculate. It is somewhat a chicken and egg problem because we want to know the class probabilities (priors) to estimate the class of an instance but we do not have the priors and should estimate them. Usually, the prior of the k -th class is estimated according to the sample size of the k -th class:

$$\hat{\pi}_k = \frac{n_k}{n}, \quad (32)$$

where n_k and n are the number of training instances in the k -th class and in total, respectively. This estimation considers Bernoulli distribution for choosing every instance out of the overall training set to be in the k -th class.

The mean of the k -th class can be estimated using the Maximum Likelihood Estimation (MLE), or Method of Moments (MOM), for the mean of a Gaussian distribution:

$$\mathbb{R}^d \ni \hat{\mu}_k = \frac{1}{n_k} \sum_{i=1}^n \mathbf{x}_i \mathbb{I}(\mathcal{C}(\mathbf{x}_i) = k), \quad (33)$$